Literature-Implied Answer: Fully $(2; 2, \ldots, 2)$-Summing L_∞ Multilinear Forms

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Source Question

The source paper is Daniel M. Pellegrino, *Cotype and nonlinear absolutely summing mappings*, arXiv:math/0307311. In Section 4, after defining r-fully absolutely summing multilinear mappings, the author asks whether every scalar n-linear mapping on $\mathcal{L}_\infty$ spaces is fully $(2; 2, \ldots, 2)$-summing. The question appears on page 10 of the source PDF.

In the source notation, the case $r = n$ is called "fully" summing, while the subsequent corollary gives an $f(2)$-fully conclusion. Thus the source paper only gives the all-indices answer in the bilinear case; for $n \geq 3$, the natural question is the full multiple-array assertion.

Supporting Theorem

The supporting paper is Geraldo Botelho, Carsten Michels, and Daniel Pellegrino, *New inclusion and coincidence theorems for summing multilinear mappings*, arXiv:0810.0947.

Section 4 of the supporting paper defines multiple $(q; p_1, \ldots, p_n)$-summing mappings by summing over all indices $(j_1, \ldots, j_n)$. This is the same all-indices condition as the source paper's fully summing notion. In Section 4.3, Corollary 4.10, page 9, the supporting paper proves:

$$\mathcal{L}(E_1, \ldots, E_n; F) = \mathcal{L}_{ms,r}(E_1, \ldots, E_n; F) \qquad (2 \leq r \leq \infty)$$

whenever $E_1, \ldots, E_n$ are $\mathcal{L}_\infty$-spaces and F has cotype 2.

Identification

The scalar field $\mathbb{K}$ has cotype 2. Applying Corollary 4.10 with $F = \mathbb{K}$ and $r = 2$ gives

$$\mathcal{L}(E_1, \ldots, E_n; \mathbb{K}) = \mathcal{L}_{ms,2}(E_1, \ldots, E_n; \mathbb{K}),$$

that is, every scalar-valued continuous n-linear form on $\mathcal{L}_\infty$ domains is multiple $(2; 2, \ldots, 2)$-summing. Under the notation match above, this is precisely the fully $(2; 2, \ldots, 2)$ answer requested in the source question.

Provenance and Scope

This packet records a literature-implied answer, not an original proof. The supporting paper does not explicitly present Corollary 4.10 as resolving the source sentence, but it cites the source paper and proves a stronger coincidence theorem whose scalar $r = 2$ case answers the question. No broader claim is made about optimal constants, other exponents, or unrelated nonlinear summability problems.

References

- D. M. Pellegrino, *Cotype and nonlinear absolutely summing mappings*, arXiv:math/0307311.
- G. Botelho, C. Michels, and D. Pellegrino, *New inclusion and coincidence theorems for summing multilinear mappings*, arXiv:0810.0947.